

Systems Biology: Mathematics for Biologists



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Last time: System of differential equations

- 2D system: $\begin{cases} \frac{dx}{dt} = f(x, y) \\ \frac{dy}{dt} = g(x, y) \end{cases}$
- Solution for linear system $\begin{pmatrix} \frac{dx}{dt} \\ \frac{dy}{dt} \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$:
 $x(t) = C_1 v_{1x} e^{\lambda_1 t} + C_2 v_{2x} e^{\lambda_2 t}$, $y(t) = C_1 v_{1y} e^{\lambda_1 t} + C_2 v_{2y} e^{\lambda_2 t}$.
- Solutions often unknown for non-linear systems.
- Qualitative analysis can tell us long-term behaviour.
- A **phase portrait** tells us where x and y increase, decrease and do not change.
- An **x null-cline** is a line on which $\frac{dx}{dt} = 0$.
- A **y null-cline** is a line on which $\frac{dy}{dt} = 0$.
- An **equilibrium** occurs where $\frac{dx}{dt} = 0$ **and** $\frac{dy}{dt} = 0 \implies$.
- The **vectorfield** shows the direction of change for x and y .

Global plan

General plan of qualitative analysis:

$$\begin{cases} \frac{dx}{dt} = f(x, y) \\ \frac{dy}{dt} = g(x, y) \end{cases}$$

- ① Draw $\frac{dx}{dt} = f(x, y) = 0$ and $\frac{dy}{dt} = g(x, y) = 0$ null-clines.
- ② Choose a region of the x, y plane and find x and y direction vectors $\vec{v} = (f(x, y), g(x, y))$.
- ③ Find the vectorfield in the adjacent regions using:
 - Flip the horizontal component when crossing x null-cline.
 - Flip the vertical component when crossing y null-cline.
- ④ Show the direction of the vectorfield on the null-clines.
- ⑤ Try determine the dynamics around an equilibrium.
- ⑥ Try to derive the global dynamics of the system.

Equilibrium types in 2D systems

Equilibrium types

1D systems:

- Two regions, left and right of an equilibrium.
- Arrows can point away or toward the equilibrium.
- So two equilibrium types possible: stable and unstable.

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2D systems:

- Four regions around an equilibrium point.
- Arrows can point away or toward the equilibrium, *or both!*
- Six different equilibrium types possible, two of which are stable.

Equilibrium types

2D systems:

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Equilibrium types can be found from:

- Eigenvalues (only works for linear systems)
$$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = C_1 \begin{pmatrix} v_{1x} \\ v_{1y} \end{pmatrix} e^{\lambda_1 t} + C_2 \begin{pmatrix} v_{2x} \\ v_{2y} \end{pmatrix} e^{\lambda_2 t}$$

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- Vectorfield (works for most but not all situations).

Equilibrium types: stable node

$$\begin{cases} \frac{dx}{dt} = -2x + y \\ \frac{dy}{dt} = x - 2y \end{cases}$$

Null-clines:

$$y = 2x$$

$$y = \frac{1}{2}x$$

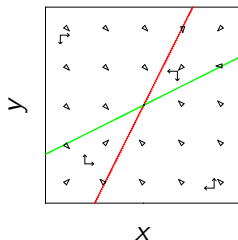
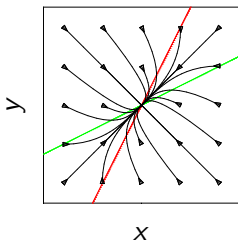
Eigenvalues:

$$\lambda_1 = -1$$

$$\lambda_2 = -3$$

Eigenvectors:

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$



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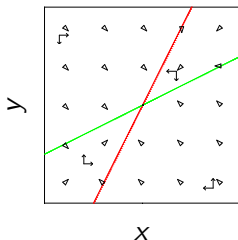
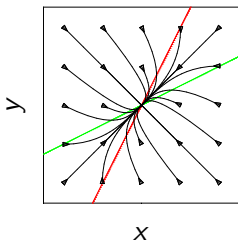
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Eigenvalues: both are negative $\rightarrow$ **stable node**

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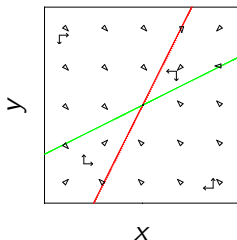
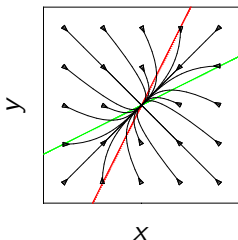
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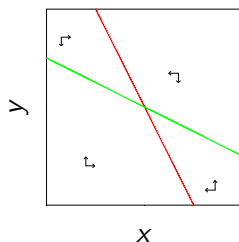


Eigenvalues: both are negative $\rightarrow$ **stable node**

Vectorfield: all arrows point to equilibrium $\rightarrow$ **stable node**

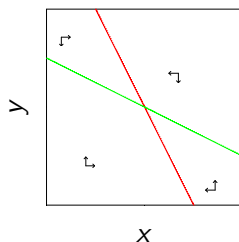
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$$\begin{cases} \frac{dx}{dt} = -2x - y \\ \frac{dy}{dt} = -x - 2y \end{cases} \quad \begin{array}{l} \text{Null-clines:} \\ y = -2x \\ y = -\frac{1}{2}x \end{array}$$



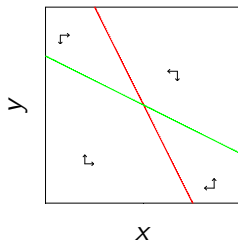
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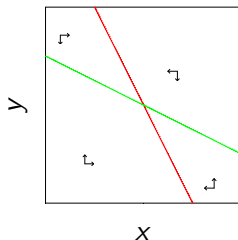
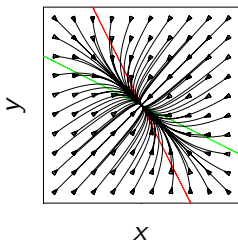
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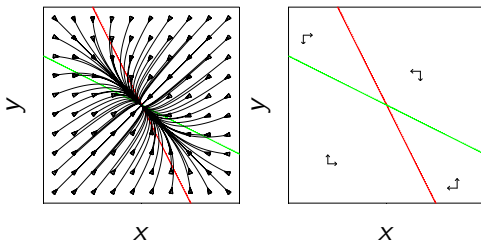
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Eigenvalues: both are negative $\rightarrow$ **stable node**

Vectorfield: all arrows point to equilibrium $\rightarrow$ **stable node**

Equilibrium types: unstable node

$$\begin{cases} \frac{dx}{dt} = 2x + y \\ \frac{dy}{dt} = x + 2y \end{cases}$$

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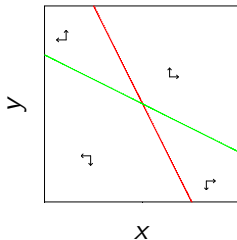
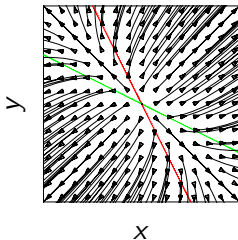
Eigenvalues:

$$\lambda_1 = 1$$

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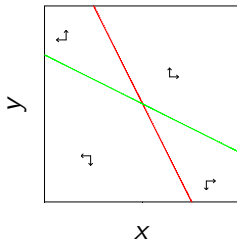
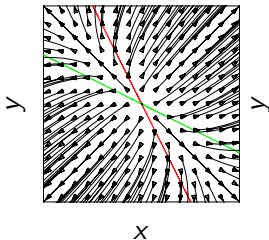
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Eigenvalues: both are positive $\rightarrow$ **unstable node**

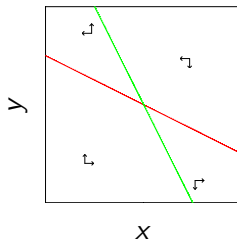
Vectorfield: all arrows point away from equilibrium $\rightarrow$ **unstable node**

Equilibrium types: saddle point

$$\begin{cases} \frac{dx}{dt} = -x - 2y \\ \frac{dy}{dt} = -2x - y \end{cases}$$

Null-clines:

$$\begin{aligned} y &= -\frac{1}{2}x \\ y &= -2x \end{aligned}$$



Equilibrium types: saddle point

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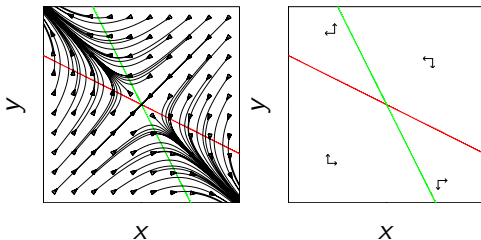
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Eigenvalues:

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Eigenvectors:

$$\begin{pmatrix} -1 \\ 1 \end{pmatrix} \quad \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$



Eigenvalues: one is negative, the other positive $\rightarrow$ **saddle point**

Vectorfield: one direction toward, one away from equilibrium $\rightarrow$ **saddle point**

When things get complex...

$$\begin{cases} \frac{dx}{dt} = -x + 2y \\ \frac{dy}{dt} = -2x - y \end{cases} \quad \begin{array}{l} \text{Null-clines:} \\ y = \frac{1}{2}x \\ y = -2x \end{array}$$

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$$(a - \lambda)(d - \lambda) - cb = (-1 - \lambda)(-1 - \lambda) - (-2 \times 2) = 0$$

$$1 + \lambda + \lambda + \lambda^2 + 4 = 0$$

$$\lambda^2 + 2\lambda + 5 = 0$$

$$\lambda_{1,2} = \frac{-2 \pm \sqrt{2^2 - 4 \times 1 \times 5}}{2}$$

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$$\mathbf{v}_{1,2} = \begin{pmatrix} -b \\ a - \lambda_{1,2} \end{pmatrix}$$

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$$\mathbf{v}_{1,2} = \begin{pmatrix} -b \\ a - \lambda_{1,2} \end{pmatrix} = \begin{pmatrix} -2 \\ -1 - (-1 \pm 2i) \end{pmatrix}$$

General complex numbers

Complex number: $z = \alpha \pm i\beta$

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General complex numbers

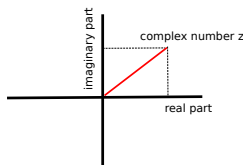
Complex number: $z = \alpha \pm i\beta$

Real part: $\text{Re}z = \alpha$

Imaginary part: $\text{Im}z = i\beta$

Can be displayed on the complex plane:

The complex plane



Looks similar to a 2D vector $\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$.

Absolute value or modulus: $|z| = \sqrt{\alpha^2 + \beta^2}$

Calculus with complex numbers

Very similar to working with vectors:

Addition of two complex numbers:

$$z_1 = 3 + 10i, z_2 = -5 + 4i$$

$$z_1 + z_2$$

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$$z_1 + z_2 = (3 + 10i) + (-5 + 4i)$$

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Addition of two complex numbers:

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$$\begin{pmatrix} 3 \\ 10 \end{pmatrix} + \begin{pmatrix} -5 \\ 4 \end{pmatrix} = \begin{pmatrix} -2 \\ 14 \end{pmatrix}$$

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Multiplication by real number:

$$z_1 = 3 + 10i$$

$$10 \times z_1$$

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$$10 \times z_1 = 10(3 + 10i)$$

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$$z_1 = 3 + 10i$$

$$10 \times z_1 = 10(3 + 10i) = 30 + 100i$$

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Multiplication by real number:

$$z_1 = 3 + 10i$$

$$10 \times z_1 = 10(3 + 10i) = 30 + 100i$$

$$10 \begin{pmatrix} 3 \\ 10 \end{pmatrix} = \begin{pmatrix} 30 \\ 100 \end{pmatrix}$$

Calculus with complex numbers

Dissimilar to working with vectors:

Multiplication of two complex numbers:

$$z_1 = 3 + 10i, z_2 = -5 + 4i$$

$$z_1 \times z_2$$

Calculus with complex numbers

Dissimilar to working with vectors:

Multiplication of two complex numbers:

$$z_1 = 3 + 10i, z_2 = -5 + 4i$$

$$z_1 \times z_2$$

$$(3 + 10i)(-5 + 4i)$$

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Calculus with complex numbers

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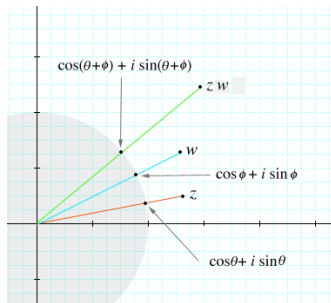
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Multiplication of two complex numbers



Complex eigenvectors

$$\begin{pmatrix} \frac{dx}{dt} \\ \frac{dy}{dt} \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \quad \lambda_{1,2} = \alpha \pm i\beta$$

Because eigenvalues are complex so are the eigenvectors:

$$\mathbf{v}_1 = \begin{pmatrix} -b \\ a - \lambda_1 \end{pmatrix} = \begin{pmatrix} -b \\ a - (\alpha + i\beta) \end{pmatrix}$$

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The real part of this eigenvector is $\mathbf{v}_{1r}$, the imaginary part is $i \times \mathbf{v}_{1i}$.

$$\mathbf{v}_{1,2} = \begin{pmatrix} -b \\ a - \alpha \end{pmatrix} \pm i \begin{pmatrix} 0 \\ \beta \end{pmatrix}$$

Complex eigenvectors

$$\begin{pmatrix} \frac{dx}{dt} \\ \frac{dy}{dt} \end{pmatrix} = \begin{pmatrix} -1 & 2 \\ -2 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \quad \lambda_{1,2} = -1 \pm 2i$$

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What does it all mean?

For a linear system

$$\begin{pmatrix} \frac{dx}{dt} \\ \frac{dy}{dt} \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

the general solution follows from the eigenvalues and -vectors:

$$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = C_1 \mathbf{v}_1 e^{\lambda_1 t} + C_2 \mathbf{v}_2 e^{\lambda_2 t}$$

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For complex eigenvalues $\lambda_{1,2} = \alpha \pm i\beta$, we get:

$$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = C_1 (\mathbf{v}_{1r} + i\mathbf{v}_{1i}) e^{(\alpha+i\beta)t} + C_2 (\mathbf{v}_{2r} + i\mathbf{v}_{2i}) e^{(\alpha-i\beta)t}$$

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But how can we tell the behavior of the system from this?

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But how can we tell the behavior of the system from this?

We can rewrite the general solution using **Euler's formula**:

$$e^{ix} = \cos x + i \sin x$$

Complex solutions

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So we can rewrite the general solution as:

$$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = C_1(\mathbf{v}_{1r} + i\mathbf{v}_{1i})e^{\alpha t}(\cos \beta t + i \sin \beta t) + C_2(\mathbf{v}_{2r} + i\mathbf{v}_{2i})e^{\alpha t}(\cos \beta t - i \sin \beta t)$$

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or even:

$$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = e^{\alpha t} \{ C_1(\mathbf{v}_{1r} + i\mathbf{v}_{1i})(\cos \beta t + i \sin \beta t) + C_2(\mathbf{v}_{2r} + i\mathbf{v}_{2i})(\cos \beta t - i \sin \beta t) \}$$

Complex solutions

What does this general solution tell us?

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The behaviour of $x(t)$ and $y(t)$ is given by an **exponential function** $e^{\alpha t}$, multiplied with a **sum of sine and cosine functions**.

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The values of x and y will show oscillatory dynamics over time.

Complex solutions

What does this general solution tell us?

$$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = e^{\alpha t} \{ C_1(\mathbf{v}_{1r} + i\mathbf{v}_{1i})(\cos \beta t + i \sin \beta t) + C_2(\mathbf{v}_{2r} + i\mathbf{v}_{2i})(\cos \beta t - i \sin \beta t) \}$$

The behaviour of $x(t)$ and $y(t)$ is given by an **exponential function** $e^{\alpha t}$, multiplied with a **sum of sine and cosine functions**.

The values of x and y will show oscillatory dynamics over time. Over time, the amplitude of the oscillations will increase (when $\alpha > 0$) or decrease (when $\alpha < 0$).

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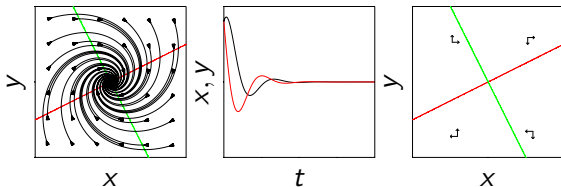
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So, in the x, y phase plane:

x and y will show outward ($\alpha > 0$) or inward ($\alpha < 0$) spiraling trajectories.

Equilibrium types: stable spiral

$$\begin{cases} \frac{dx}{dt} = -x + 2y \\ \frac{dy}{dt} = -2x - y \end{cases} \quad \begin{array}{l} \text{Null-clines:} \\ y = \frac{1}{2}x \\ y = -2x \end{array} \quad \begin{array}{l} \text{Eigenvalues:} \\ \lambda_1 = -1 + 2i \\ \lambda_2 = -1 - 2i \end{array} \quad \begin{array}{l} \text{Eigenvectors:} \\ \begin{pmatrix} i \\ 1 \end{pmatrix} \quad \begin{pmatrix} i \\ -1 \end{pmatrix} \end{array}$$



Eigenvalues: complex with negative real part $\rightarrow$ **stable spiral**

Vectorfield: arrows only suggest rotation

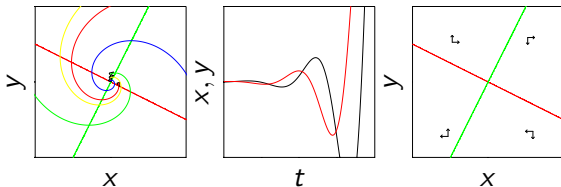
Equilibrium types: unstable spiral

$$\begin{cases} \frac{dx}{dt} = x + 2y \\ \frac{dy}{dt} = -2x + y \end{cases}$$

Null-clines: $y = -\frac{1}{2}x$
 $y = 2x$

Eigenvalues: $\lambda_1 = 1 + 2i$
 $\lambda_2 = 1 - 2i$

Eigenvectors: $\begin{pmatrix} -i \\ 1 \end{pmatrix}$ $\begin{pmatrix} i \\ 1 \end{pmatrix}$

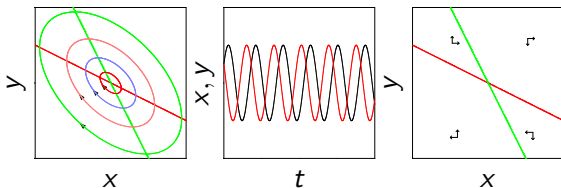


Eigenvalues: complex with positive real part $\rightarrow$ **unstable spiral**

Vectorfield: arrows only suggest rotation

Equilibrium types: center point

$$\begin{cases} \frac{dx}{dt} = x + 2y \\ \frac{dy}{dt} = -2x - y \end{cases} \quad \begin{array}{l} \text{Null-clines:} \\ y = -\frac{1}{2}x \\ y = -2x \end{array} \quad \begin{array}{l} \text{Eigenvalues:} \\ \lambda_1 = i\sqrt{3} \\ \lambda_2 = -i\sqrt{3} \end{array} \quad \begin{array}{l} \text{Eigenvectors:} \\ \begin{pmatrix} -\frac{1}{2} \pm \frac{i\sqrt{3}}{2} \\ 1 \end{pmatrix} \end{array}$$

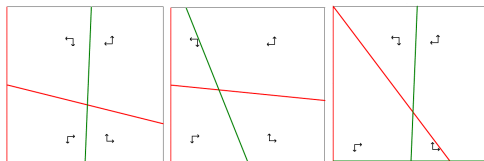


Eigenvalues: complex with zero real part $\rightarrow$ **center point**

Vectorfield: arrows only suggest rotation

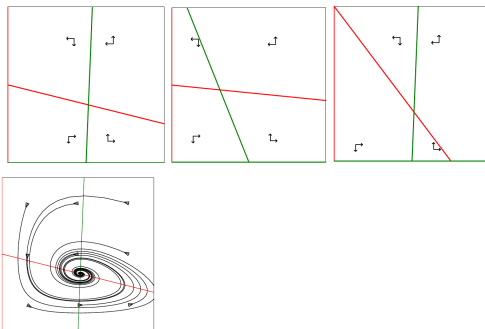
Vectorfield insufficient

Sometimes the vectorfield does not give enough information:



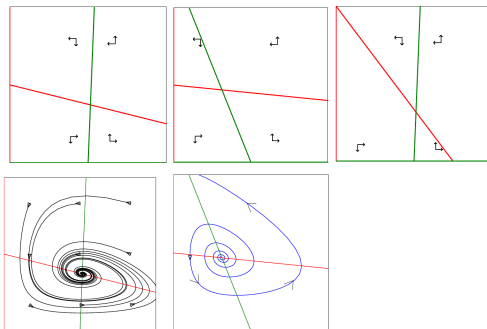
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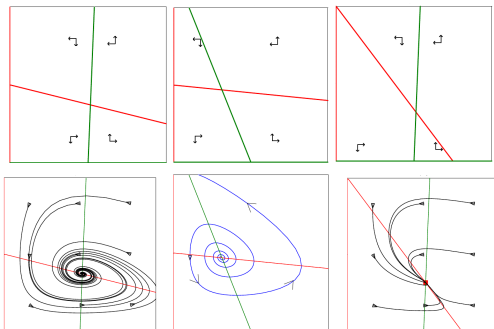
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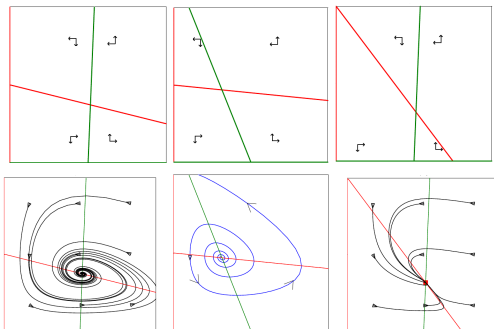
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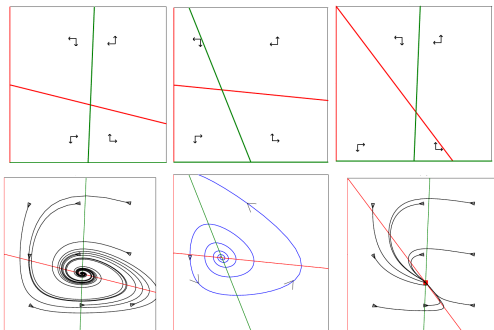
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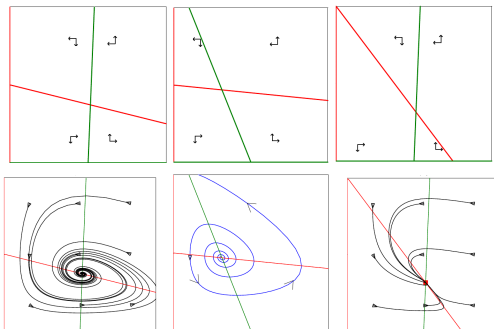


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All these vectorfields suggest rotation, but we may not even have a spiral!

If all feedback is negative $\rightarrow$ equilibrium is stable, but type unknown!

If one of feedbacks is positive $\rightarrow$ equilibrium is an unstable spiral!

An overview of 2D equilibria

An equilibrium is only stable, if from *all* directions trajectories converge on it. One or more diverging directions means that the equilibrium is unstable.

- **Stable equilibria:**

- stable node: real, negative eigenvalues ($\lambda_1 < 0, \lambda_2 < 0$)
- stable spiral: complex eigenvalues with real part $\alpha < 0$

- **Unstable equilibria:**

- unstable node: real, positive eigenvalues ($\lambda_1 > 0, \lambda_2 > 0$)
 - saddle node: real eigenvalues, $\lambda_1 < 0$ and $\lambda_2 > 0$ or vice versa
 - unstable spiral: complex eigenvalues with real part $\alpha > 0$
- A center point is a **neutrally stable equilibrium** ($\alpha = 0$):
no convergence or divergence!

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So an equilibrium is stable if the real part of both eigenvalues is negative.